

About National Air Quality Index (AQI)

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Overview

The Air Quality Index provides a single, easy-to-understand number that communicates overall air quality and its likely health significance. It converts technical pollution measurements into a common public information format so that citizens, administrators, healthcare providers, schools, and local authorities can make informed decisions.

Why AQI was introduced

India required a harmonized public communication tool that could summarize multiple pollutants and help non-specialists understand daily air pollution conditions.

What AQI represents

AQI expresses the highest relevant pollution burden among the monitored pollutants and links it directly with practical health interpretation.

Official Purpose: AQI helps people quickly understand whether today's air is safe, cautionary, or harmful. It supports public awareness, health advisories, emergency planning, and environmental governance under

continuous air quality monitoring programmes.

Citizen Use: Check the AQI before outdoor exercise, school commuting, construction scheduling, and protection of vulnerable family members.

National AQI Scale

Category	AQI Range	Health Significance
Good	0–50	Air quality poses little or no risk. Minimal health impact.
Satisfactory	51–100	Minor breathing discomfort to sensitive individuals. Generally acceptable for the broader population.
Moderate	101–200	Breathing discomfort to people with lung disease, asthma, children, and older adults. Sensitive groups should reduce prolonged exertion outdoors.
Poor	201–300	Breathing discomfort to most people on prolonged exposure. Outdoor activity should be moderated, especially for at-risk groups.
Very Poor	301–400	Respiratory illness risk increases on prolonged exposure. Children, elderly persons, and those with illness need added protection.
Severe	401–500	Significant health effects even on healthy individuals. Serious impacts on people with existing respiratory or cardiovascular diseases.

From Complex Measurements to a Single Public Number

1. **Monitoring Station Data** — Continuous stations record pollutant concentrations such as PM2.5, PM10, NO2, SO2, CO, O3, NH3, and Pb.
2. **Sub-Index Conversion** — Each measured concentration is translated into a pollutant-specific sub-index using the AQI breakpoint framework.
3. **Final AQI Value** — The highest relevant sub-index is used to represent overall air quality and trigger the associated health category.

Role of Continuous Air Quality Monitoring Stations

Continuous Ambient Air Quality Monitoring Stations provide regular measurements of urban air pollutants, enabling near real-time assessment, trend analysis, public alerts, regulatory review, and evidence-based response during episodes of elevated pollution. Each pollutant concentration is converted into a sub-index according to the notified AQI breakpoint table, and the highest relevant sub-index becomes the reported AQI category and public health signal for that period.

Decision Support Value:

- For citizens: Plan exercise, commuting, school exposure, and mask use.
- For administrators: Guide advisories, traffic measures, dust control, and emergency response.
- For health systems: Prepare communication for vulnerable groups during episodes of poor air quality.

Criteria Pollutants Used in India's AQI

Pollutant	Simple Explanation	Common Sources	Potential Health Impacts
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PM2.5 (Fine Particulate Matter)	Microscopic airborne particles that penetrate deep into the lungs.	Vehicle exhaust, biomass burning, coal combustion, diesel generators, and secondary aerosol formation.	Irritation of airways, aggravated asthma, reduced lung function, and long-term cardiovascular risk.
PM10 (Particulate Matter)	Coarser inhalable particles suspended in the ambient air.	Road dust, construction activity, quarrying, industrial handling, and re-suspended soil dust.	Coughing, throat irritation, breathing discomfort, and worsening of respiratory disease.
NO2 (Nitrogen Dioxide)	A reactive gas formed mainly during high-temperature fuel combustion.	Motor vehicles, thermal power plants, industrial boilers, and urban traffic corridors.	Airway inflammation, reduced lung defense, and greater sensitivity among asthmatics.
SO2 (Sulfur Dioxide)	A pungent gas associated with sulfur-containing fuels.	Coal-fired power generation, industrial furnaces, petroleum refining, and metal smelting.	Bronchial irritation, chest tightness, and aggravation of respiratory symptoms.

CO (Carbon Monoxide)	A colorless toxic gas generated by incomplete combustion.	Vehicle exhaust, generator sets, brick kilns, and poorly controlled combustion sources.	Reduces oxygen delivery in the body, causing headache, dizziness, and stress on the heart.
O3 (Ozone)	A secondary pollutant formed by sunlight-driven reactions in urban air.	Reaction of NOx and VOC emissions from traffic, solvents, and industrial sources.	Eye irritation, chest discomfort, breathing difficulty, and reduced outdoor work capacity.
NH3 (Ammonia)	A reactive alkaline gas that also contributes to particulate formation.	Waste handling, fertilizers, agricultural activity, sewage, and some industrial processes.	Irritation of eyes and throat and a role in forming harmful secondary particulates.
Pb (Lead)	A toxic heavy metal present in some industrial air emissions and contaminated dust.	Battery recycling, metal processing, industrial stacks, and contaminated suspended dust.	Neurological harm, developmental risk in children, and broader chronic toxicity.

Common Urban Air Pollution Sources

- **Vehicle Emissions** — Tailpipe exhaust from cars, buses, trucks, and two-wheelers contributes particulate matter, NO₂, CO, and ozone precursors.
- **Industrial Emissions** — Stacks from industries, boilers, and thermal sources emit sulfur dioxide, nitrogen oxides, and

particulates.

- **Construction Dust** — Demolition, excavation, road cutting, and movement of materials increase PM10 and local dust burdens.
- **Biomass Burning** — Burning of crop residue, waste, wood, and solid fuel releases PM2.5, CO, and toxic gases.
- **Urban Mixed Sources** — Dense settlements combine traffic, energy use, dust, open burning, and commercial activity into cumulative exposure.

How Citizens Can Use AQI Information

Daily Practical Use:

- Limit outdoor activity during high AQI periods, especially strenuous exercise.
- Use masks when pollution is severe or when travel during high pollution cannot be avoided.
- Protect children, elderly individuals, and people with respiratory or cardiovascular diseases.
- Check daily AQI reports and health advisories before planning school, travel, or outdoor work.

Health Advisory Focus:

AQI Level	Suggested Public Action
Good to Satisfactory	Normal activity can continue with routine awareness.
Moderate to Poor	Sensitive persons should reduce prolonged outdoor exertion and monitor symptoms.
Very Poor	Limit outdoor exposure, protect vulnerable groups, and follow local advisories.
Severe	Avoid unnecessary outdoor activity, consider masks, and reduce exposure immediately.

Public Message: Clean air information should be usable, timely, and linked with simple protective action.

National Air Quality Index information for public awareness and educational use. (CPCB / NAQI / NAMP / AQI)